Beyond Eventual Consistency

TM

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Problem Space

I prefer tools to writing more code

Cassandra

Wonderful database for storing large amounts of data

Cassandra

Doesn’t solve all of my business problems

CAP Theorem

Defined how we’ve thought about data for years

Do new theories disprove it?

Are we optimizing for the wrong case?

Are failures frequent enough that we have to overcompensate?

Downtime

Does anyone have a database that never goes down?

PACELC, by Daniel Abadi

if there is a partition (P), how does the system trade off availability and consistency (A and C); else (E), when the system is running normally in the absence of partitions, how does the system trade off latency (L) and consistency (C)?

Data Management Taxonomy

Consistency

ACID versus CAP

Serializability

Can I have guaranteed write order?

Linearizability

All nodes agree on order

Linearizable Consistency

Will I always get the most recent value?

Optimistic versus Pessimistic

Do I lock or not?

Solutions

There are more than you think

Amazon Aurora

mysql in the Cloud

Amazon Aurora

- mysql in the cloud, but four 9s of availability
- Compatible with existing solutions (JDBC, etc)
- One single write master with eventually consistent replicas
- Has quorum read/write but really redundant writes across AZs
- Performance limit to largest deployed single instance
- Throughput at the cost of latency (pessimistic locking)
- Vertical scalability
- No sharding

Microsoft Cosmos

Declarative consistency

Attend Rimma Nehme's talk in Room 3 today at 13:35

Azure Cosmos

- Global database infrastructure as a service
- ACID style transactional applications
- Evolved from DocumentDB, can act like a Mongo store
- Can now do graphs and KV
- Declarative consistency model, from eventual to strong
- No uniqueness or foreign key constraints
- No distributed transactions or GroupBy (yet)

Google Cloud Spanner

Global consistency

Google Spanner

- Globally distributed consistent data
- Only on the GCP
- Uses TrueTime approach for snapshots and synchronization
- Not compatible with existing database solutions
- Utilizes pessimistic locking, supported by Google's WAN
- Five 9s availability with strong consistency
- Spanner and Cosmos offer the same guarantee for throughput, latency and consistency

CockroachDB

The FOSS solution

CockroachDB

- OSS solution for distributed storage using Spanner’s approach
- Can’t use TrueTime
- Relies on a highly-latent approach leveraging NTP
- Passed the Jepsen tests

FaunaDB

Cloud agnostic consistency

FaunaDB - things to know

- PACELC, based on Calvin paper
- Transactional, with single-phase commit, across thousands of records and indices and optimistic approach
- Serialized, uses Raft to manage order of updates
- Linearly consistent
- Hierarchical row-level authorization for tenants and workloads
- Supports multiple indices
- Supports temporality, good for audit, rollback and cache coherency
- Supports graphs

FaunaDB - things to know

- Closed source
- Transactions are limited to a single function
- Has its own somewhat limited type-safe, functional query language
- It’s a new product and growing in maturity
- Not many existing users
- Not verified by Jepsen
- What if it disappears? (FoundationDB, RethinkDB, Riak)

Cassandra

- Let’s watch what DataStax does next to compete in this space

Thank you!